

Renewable Security: India's Next Strategic Advantage



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Every once in a while, a country reaches a moment where technology, ambition, and necessity intersect so sharply that the future stops being a distant idea and becomes a blueprint waiting to be built. India is at that moment - on the cusp of a 500-gigawatt renewable ambition by 2030 - a target that is not simply about numbers, but about resilience, industrial competitiveness, and national security.

In that time, ENGIE has grown from its first solar project to a 2.68-gigawatt portfolio spread across 7 states and 24 projects. Our next horizon is clear: scaling to 7 gigawatts by 2030. This journey mirrors India's own evolution — from early adoption to becoming one of the most decisive actors in the global energy transition.

Energy Security as Economic Strategy

India's peak power demand has already crossed 250 GW and is set to nearly double by 2032. Behind this surge lies not just households and factories, but the demands of data centres, electric mobility, AI infrastructure, and green hydrogen. The real question is not whether India can meet this demand, but how: with a fragile fossil-heavy base, or with clean, reliable power that anchors growth.

Energy security is therefore no longer a side note in climate strategy. It is the central determinant of India's economic resilience and its global influence. Every additional megawatt is not just electricity; it is a building block of industrial competitiveness, urban futures, and geopolitical weight. This is where hybrid systems and storage-backed architectures become non-negotiable. The future backbone of India's grid will not be standalone solar or wind — it will be integrated renewable ecosystems.



From Capacity to Reliability: The Hybrid Imperative

India now ranks fourth globally in installed renewable capacity, with over 80 GW added in the past five years. Yet intermittency remains the Achilles' heel. Reliability is the foundation of economic competitiveness — and this is where hybrid systems of solar, wind, and battery storage become indispensable. The Ministry of Power's new mandate requiring co-located storage for future solar tenders reinforces this shift.

In Gujarat and Rajasthan, we already see investment moving decisively toward solar-wind-storage hybrids. Recently we won the first BESS stand alone project with 280 MW/ 2 hrs of storage capacity. This is our second largest battery project in the world in terms of installed capacity, contributing to our ambition to be a leading player in large-scale battery storage systems. As of June 30, 2025, the Group's portfolio of BESS projects in operation or under construction amounted to 5.6 GW worldwide.

A Business Model for Scale

India's renewable expansion will stand or fall on the strength of its financing and offtake models. Utility-scale Power Purchase Agreements (PPAs) with SECI and NTPC remain foundational, but the fastest growth is coming from corporates. Data centres, cement, and steel producers are turning to Corporate PPAs and Virtual PPAs (VPPAs) to secure price certainty and sustainability credentials.

From 12 GW in 2023, the corporate procurement market is projected to reach 100 GW by 2030. ENGIE is leveraging 4.3 GW of global corporate PPA experience to deliver storage-backed, tailor-made solutions for Indian corporates. Through our Catalyst Connect industry dialogue series, we have also convened policymakers, regulators, and financiers to address bottlenecks in transmission, VPPA readiness, and payment security.